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Maths puzzles for early finishers

Early Finisher Puzzles

Primary 4

Semester 1

Terms 1 and 2

20 single-page puzzles
Almost every puzzle checks itself
One star and two star pages
Teacher answer key included

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Notes

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Teacher notes (DRAFT)

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What this is

Twenty single-page maths puzzles for Primary 4 pupils who finish their work early. This is Semester 1 of two booklets (Terms 1 and 2). Each page stands alone, so you can hand out one page or keep a tray of pages. The pupil needs no help to start.

Which terms

The pages are grouped by the order in which Primary 4 topics are typically taught: this booklet follows Terms 1 and 2, the other booklet follows Semester 2 (Terms 3 and 4). Schools vary, so check your scheme of work: use a page whenever its topic has been taught, from either booklet.

Stars

One star pages practise a topic as it is usually taught (10 pages). Two star pages use the same topics with two steps or two clues to combine (10 pages).

How the pages check themselves

The boxed strip at the foot of each page tells the pupil how to check. The check shows that something is wrong. It does not give the answer, and it asks the pupil to see whether the answer makes sense. Every answer box and every letter box must be filled: a box left blank gives no letter, and the check cannot see it.

- **Code word.** The answers give letters that spell a word. The strip gives no hint of the word. Each word can be spelt from its letters in one way only, so answers in the wrong boxes never make another word, and one or two wrong answers very rarely do. A box left empty gives no letter and is not seen by the check.
- **Colour by code.** Each answer is found in a key of five numbers and the part is coloured with that crayon. The pupil counts the parts of each colour and compares with the box on the page; matching parts must share a colour.
- **Route.** The line must reach FINISH through the stated number of boxes, and every box on it must follow the rule on the page.
- **Loop.** Following the arrows round the ring, the last arrow must bring the pupil back to the start number.

Marking is not needed. To spot-check, look only at the check strip.

How the checks were tested

A script tried every single wrong answer and every wrong arrangement on every page. Each check rejected at least 95% of them. A wrong answer that spells any of about 241 000 English words counts as not rejected. Two wrong answers at once were not all tested.

Teacher notes: using the pages (DRAFT)

Using the pages

- **Time.** Most pages take 5 to 15 minutes; the colour-by-code pages (S1-05, S1-11) take longer because pupils colour a picture. Pupils need their own crayons (five colours: yellow, red, blue, green, orange). For a pencil-only copy the pupil writes Y, R, B, G or O in each part; the tally and the matching parts are checked the same way. Light crayons work best: a dark crayon hides the number in the part on a photocopy.
- **If the check fails.** The pupil goes back, looks for the slip, and asks a friend or you if still stuck.
- **Problem solving.** Some pages (S1-08, S1-09, S1-12, S1-13, S1-16, S1-20) practise the kinds of question that appear in school tests, such as stories with more than one step, remainders and unequal sharing, rather than one numbered syllabus outcome. The key tags them problem solving.
- **Stars.** One star: one skill, as usually taught. Two stars: the same topics with two steps or two clues to combine.
- **Drawings.** Pictures of figures, solids, nets, graphs and protractors are original drawings. Rectilinear figures are drawn to a scale, but pupils are never asked to measure the page. A protractor picture is for reading only. Pupils use their own protractor and ruler only if a page says so.

Photocopying

All pages are black and white with solid lines, so they copy cleanly. Use normal toner, not a light setting, so that the thin dashed lines on the pie charts and graphs stay visible. Each page is one side only. Pupils write on the page, and the answer boxes are large. Use A4 and print at 100% (actual size).

Pictures are not actual size.

Contents: Semester 1

Page	Code	Puzzle	Tier
S1-01	P4-01	Place Value	1 star
S1-02	P4-02	Words and Numerals	1 star
S1-03	P4-03	Number Patterns	2 stars
S1-04	P4-04	Round It	1 star
S1-05	P4-05	Colour the Bus	2 stars
S1-06	P4-06	Factor or Not?	1 star
S1-07	P4-07	Multiples Maze	1 star
S1-08	P4-08	Common Factors	2 stars
S1-09	P4-09	Number Riddles	2 stars
S1-10	P4-10	Long Multiplication	1 star
S1-11	P4-11	Colour the Lorry	2 stars
S1-12	P4-12	Division	2 stars
S1-13	P4-13	Number Ring	2 stars
S1-14	P4-14	Mixed or Improper	1 star
S1-15	P4-15	Fraction Number Line	1 star
S1-16	P4-16	Fraction of a Set	1 star
S1-17	P4-17	Fraction Sums	2 stars
S1-18	P4-18	Name the Angle	1 star
S1-19	P4-19	Protractor Readings	2 stars
S1-20	P4-20	Fraction Stories	2 stars

Pages are independent. The footer code shows the page in this booklet and, in brackets, its number in the full set of 40 (P4-01 to P4-40).

What the booklet covers

Numbers up to 100 000 1.1 number notation, representations and place values (ten thousands, thousands, hundreds, tens, ones) **S1-01, S1-05**

Numbers up to 100 000 1.2 reading and writing numbers in numerals and in words **S1-01, S1-02, S1-05**

Numbers up to 100 000 1.3 comparing and ordering numbers **S1-05**

Numbers up to 100 000 1.4 patterns in number sequences **S1-03, S1-05**

Numbers up to 100 000 1.5 rounding numbers to the nearest 10, 100 or 1000 **S1-04, S1-05**

Numbers up to 100 000 1.6 use of ≈ **S1-04**

Factors and Multiples 2.1 factors, multiples and their relationship **S1-06, S1-07, S1-08, S1-09**

Factors and Multiples 2.2 determining if a 1-digit number is a factor of a given number within 100 **S1-06**

Factors and Multiples 2.3 finding the common factors of two given numbers **S1-08, S1-09**

Factors and Multiples 2.4 determining if a number is a multiple of a given 1-digit number **S1-06, S1-07, S1-09**

Factors and Multiples 2.5 finding the common multiples of two given 1-digit numbers **S1-09**

Four Operations 3.1 multiplication algorithm (up to 4 digits by 1 digit; up to 3 digits by 2 digits) **S1-10, S1-11, S1-13**

Four Operations 3.2 division algorithm (up to 4 digits by 1 digit) **S1-11, S1-12, S1-13**

Mixed Numbers and Improper Fractions 1.1 mixed numbers, improper fractions and their relationship **S1-14, S1-15, S1-17**

Fraction of a Set 2.1 fraction as part of a set **S1-16, S1-20**

Fractions: Addition and Subtraction 3.1 adding and subtracting fractions with denominators of given fractions not exceeding 12 and not more than two different denominators **S1-17, S1-20**

Angles 1.1 using notation such as ∠ABC and ∠a to name angles **S1-18**

Angles 1.2 measuring angles in degrees **S1-19**

Practised in the other booklet: Decimals up to 3 decimal places 1.1, 1.2, 1.3, 1.4, 1.5; Decimals: Addition and Subtraction 2.1; Decimals: Multiplication and Division 3.1, 3.2, 3.3; Area and Perimeter 1.1, 1.2, 1.3; Rectangle and Square 2.1, 2.2; Line Symmetry 3.1, 3.2, 3.3; Nets 4.1, 4.3, 4.4; Tables, Line Graphs and Pie Charts 1.1, 1.2.

No page: drawing an angle of a given size (Angles 1.3) and drawing a solid (Nets 4.2). A printed page cannot check a drawing, so these are left to your lessons.